

IMPORTANT TOPICS IN ROBOTICS AND INTELLIGENT SYSTEMS (AIT304)

MODULE I: ROBOT CONFIGURATIONS AND CHARACTERISTICS

High-Priority Topics:

1. **Robot Configurations** *(Repeated Questions)*
 - Four basic robot configurations according to coordinate systems
 - Cylindrical and spherical configurations
 - Work envelope of robots
2. **Robot Features and Dynamics** *(Repeated Questions)*
 - General features of wheeled, legged, and aerial robots
 - Dynamic characteristics of robots and mobile robots
3. **Grippers and End Effectors**
 - Types of grippers used in robot manipulators
 - Gripper force calculations
 - Operation principles of various grippers
4. **Robot Laws and Applications**
 - Three laws of robotics
 - Applications of robots
5. **Robot Anatomy**
 - Joint configurations
 - Degrees of freedom
 - Arm and body motion

MODULE II: ACTUATORS AND SENSORS

High-Priority Topics:

1. **Sensor Characteristics** *(Repeated Questions)*
 - Dynamic range, linearity, and resolution of sensors
 - Sensor vs. transducer differences
 - Sensor selection criteria for robotic applications
2. **Vision Sensors** *(Repeated Questions)*
 - CCD vs. CMOS camera comparison
 - Camera sensor hardware interfacing

3. **Position and Motion Sensors** *(Repeated Questions)*

- Hall effect sensor working principle and applications
- Position and velocity internal sensors
- Touch sensors

4. **Motors and Actuators** *(Repeated Questions)*

- Stepper Motor vs. Servomotor comparison
- DC motor control using H-Bridge and PWM
- Permanent magnet stepper motor operation

5. **Motion Control**

- PID controller applications
- Mobile robot movement control
- Control drives

MODULE III: ROBOT VISION AND KINEMATICS

High-Priority Topics:

1. **Robot Vision Systems** *(Highly Repeated)*

- Seven stages of robot vision/robotic vision
- Image processing analysis importance
- Pre-processing, segmentation, and recognition

2. **Holonomic vs. Non-holonomic Robots** *(Highly Repeated)*

- Characteristics and differences
- Applications of each type

3. **Robot Movement and Kinematics**

- Omnidirectional robots and configurations *(Repeated Question)*
- Differential drive significance
- Steerability and maneuverability

4. **Coordinate Frames and Transformations**

- Pure translation, pure rotation, and combined transformation
- Frame movement in space
- Mapping and transformation of coordinate systems
- Position and orientation representation

MODULE IV: LOCALIZATION AND MAPPING

High-Priority Topics:

1. **Robot Localization** *(Repeated Questions)*

- Mobile robot localization concept
- Localization problems and challenges
- Probabilistic map-based localization
- Kalman Filter method in localization

2. **SLAM (Simultaneous Localization and Mapping)** *(Repeated Questions)*

- Mathematical definition and concept
- Applications and advantages
- Graph-based vs. particle SLAM
- Integration with wheel odometer data

3. **Odometry and Error Modeling** *(Repeated Questions)*

- Accurate odometry concept
- Error sources in odometric data
- Error model for odometric position estimation

4. **Map Representation**

- Decomposition strategies
- Current challenges in map representation

MODULE V: PATH PLANNING AND CONTROL

High-Priority Topics:

1. **Path Planning Algorithms** *(Highly Repeated)*

- Dijkstra's algorithm with examples *(Most Repeated Question)*
- Bug algorithm for obstacle avoidance
- Potential field method
- Breadth-first vs. depth-first search algorithms
- Traversal algorithms comparison

2. **Control Decomposition** *(Repeated Questions)*

- Control decomposition concept
- Horizontal decomposition of robotic control systems
- Modularity for code reuse and sharing

3. **Neural Networks in Robotics**

- Neural network approach in robot navigation
- Processing sensory data (cameras, Lidar)

- Neural networks for navigation improvement

4. Obstacle Avoidance

- Bug algorithm assumptions
- Dynamic window approaches (local vs. global)

MOST FREQUENTLY REPEATED QUESTIONS ACROSS MODULES:

1. **Dijkstra's algorithm with examples** (*Module V - Asked in 4 exams*)
 - 2023 June, 2024 Dec, 2024 Jan, 2024 May
2. **Seven stages of robot vision** (*Module III - Asked in 3 exams*)
 - 2024 Dec, 2024 Jan, 2024 May
3. **Hall effect sensor working principle** (*Module II - Asked in 3 exams*)
 - 2024 Dec, 2024 Jan, 2024 May
4. **Holonomic vs. nonholonomic robots** (*Module III - Asked in 3 exams*)
 - 2024 Dec, 2024 Jan, 2024 May
5. **Dynamic range, linearity, and resolution of sensors** (*Module II - Asked in 3 exams*)
 - 2024 Dec, 2024 Jan, 2024 May
6. **Control decomposition concept** (*Module V - Asked in 3 exams*)
 - 2023 June, 2024 Dec, 2024 May
7. **Omnidirectional robot configurations** (*Module III - Asked in 3 exams*)
 - 2023 June, 2024 Dec, 2024 May
8. **Probabilistic map-based localization** (*Module IV - Asked in 2 exams*)
 - 2023 June, 2024 May
9. **Robot configurations according to coordinate systems** (*Module I - Asked in 2 exams*)
 - 2023 June, 2024 May
10. **CCD vs. CMOS Camera comparison** (*Module II - Asked in 2 exams*)
 - 2023 June, 2024 Dec